

Eliyas Abraha

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👤 PROFILE

Enthusiastic robotics engineer skilled in perception, localization, mapping, and motion planning for autonomous systems. Experienced in multi-sensor fusion and real-time robotics, with strong proficiency in Python, C++, and ROS1/2. Passionate about building robust and efficient intelligent systems.

🛠️ WORK EXPERIENCE

02/2025 – Present Enshede,Netherlands	Saxion University of Applied Sciences Smart Mechatronics and Robotics (SMART)  Robotics Engineer - Developed a multi-sensor fusion pipeline combining LiDAR, IMU, and prior map data, leveraging scan-to-map matching and high-frequency odometry. - Investigated state-of-the-art Visual Odometry, LiDAR Odometry, and Visual/LiDAR SLAM, and performed benchmarking using real-world datasets. Project LINK - Gained hands-on experience in sensor selection, calibration, and deployment of SLAM algorithms.
07/2024 – 08/2025 Böhmenkirch, Germany.	Farming Revolution GmbH  Robotics Software Engineering - Remote - Automated the processing of geospatial field data (SHP, KML) to extract field boundaries and obstacles, reducing manual path planning effort by over 90%. - Designed a row-based coverage path planning algorithm tailored to field geometry, ensuring efficient and uniform traversal for precision weeding. - Tested the algorithm in both simulated and real-world field trials, demonstrating reliable robot navigation and high-coverage accuracy.
02/2020 – 2023 Mekelle, Ethiopia	Lion International Bank  Software Engineer - Ensured reliability of database-backed systems by debugging and optimizing backend services, data pipelines, and storage layers. - Delivered IT support and digital infrastructure training to staff, increasing operational reliability and client satisfaction.

🎓 EDUCATION AND TRAINING

09/2023 – 06/2025 Spain	Erasmus Mundus Joint Master in Intelligent Field Robotic Systems (IFROS) Erasmus Mundus Joint Master in Intelligent Field Robotic Systems (IFROS)  - Studied Autonomous Systems, Probabilistic Robotics, Machine Learning, and Robot Manipulation. - Specialized in Computer Vision ,Deep Neural Network Development and 3D Sensing and Sensor Fusion. - GPA 9.1/10
09/2014 – 07/2019 Ethiopia	B.Sc. in Computer Science and Engineering Mekelle University - Algorithm Design, Data structure, Linear Algebra, AI, and Computer Network.. - GPA 3.92/4

TECHNICAL SKILLS

Programming Languages

Python, C++, MATLAB, JavaScript
(Node.js, Vue.js)

Robotics Framework

ROS1/ROS2, Gazebo,
Rviz,GTSAM,Behavioral Trees

Geospatial Libraries

Shapely, GeoPandas,QGIS

Databases Management

MongoDB and MySQL

Computer Vision and AI

OpenCV, PCL, PyTorch, YOLO

Document writing

LATEX and Microsoft Office
tools

Development Tools

Docker, Git, Linux, VS Code,
Conda

Sensors and Perception

LiDAR, GPS, Camera (RGB/RGB-
D), IMU, Sensor Fusion

OTHER LANGUAGES

English



HONOURS AND AWARDS

01/05/2019

Outstanding Performance in Huawei ICT Competition Global Final
Huawei Academy

PROJECTS

02/2024 – 05/2024

Pose Based SLAM using ICP LIDAR Scan Matching 

**Autonomous Exploration of Unknown Environments Using Dubins
Path-Oriented RRT*** 

02/2025 – 06/2025

Robust-LiDAR-Inertial-Re-Localization(ROS2 and C++) 

This thesis presents a robust and real-time localization framework for GNSS-denied environments by fusing LiDAR-Inertial Odometry (FAST-LIO2) with multithreaded NDT-based map matching using a sliding-window factor graph.

09/2025 – 02/2026

Autonomous Volume Estimation of Bulk Feed in Unstructured Environments 

This project investigates and implements perception-based methods for estimating the volume of cow feed in unstructured stable environments to enable closed-loop autonomous feeding.

09/2024 – 12/2024

Autonomous outdoor GPS waypoint navigation 

This project implements GPS waypoint navigation in an outdoor environment using Wheel Odoemtry, GPS and IMU reading. It also includes path planning and dynamic obstacle avoidance using the DWA local planner included in the move-base ros package.

09/2025 – 01/2026

Practical Guide to SLAM for Industrial Applications 

Prepared and structured a practical course on SLAM focused on real-world industrial use cases, including benchmarking and sensor calibration.

Monocular Visual Odometry with KITTI Dataset 

Stereo Vision and Object Depth Estimation 

This project presents an economical alternative to LiDAR sensors for per-pixel depth estimation in drone applications. By utilizing block matching and Semi-Global Block Matching (SGBM) algorithms, we showcase how stereo computer vision techniques can accurately determine depth information.